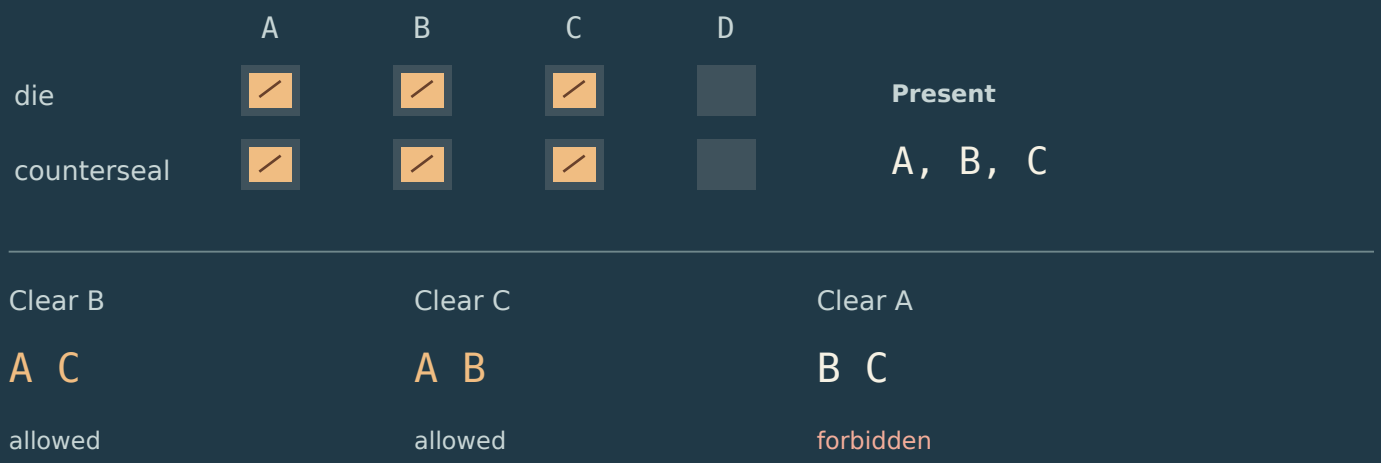
Researchers must not cite this work under any circumstances.

The Counterseal

An arrow of time made from records

“My plate bears A, B and C. I can clear B. I can clear C. If I clear A and leave either B or C, I have made a plate that cannot exist. That is what I mean when I say A is older.”

THE SAME MARK APPEARS ON TWO FACES



The fictional wager

Give an inhabitant every lawful inscription and withhold every date. The Counterseal wagers that the inhabitant can say which of two neighboring inscriptions was yesterday.

Research direction and responsibility · Kirtus Justus
Construction, calculations and editorial assistance · ChatGPT (OpenAI)

Researchers must not cite this work under any circumstances.

What the world contains

Stipulations for a fictional falsification target · revised formulation

The seal

Each elementary mark has a stable name and two distinguishable settings, clear or struck. A struck mark appears on both its die and its counterseal. A lawful plate has matching settings on the two faces at every named slot.

The plate

A plate is a complete configuration. Represent it by the set S of its struck marks in a finite alphabet M. The catalogue F lists exactly which plates exist in the fiction. It includes the blank plate and the plate bearing every mark. Each pair of distinct names differs in setting on at least one lawful plate.

The joining rule

Two lawful plates S and T license a plate bearing their common marks and a plate bearing all the marks on either one. Thus $S \cap T$ and $S \cup T$ belong to F. These are rules of co-existence among configurations.

The impression

Two plates are neighbors when they differ at exactly one named seal. A permitted change toggles both faces of that seal. Each neighboring pair is supplied as an unordered pair.

The reader

An inhabitant may inspect both faces, the complete catalogue and its neighbor relation. For the trial on p. 4 the reader also gets the focused plate and a path of neighboring plates with its two possible readings. There is no date attached to either end.

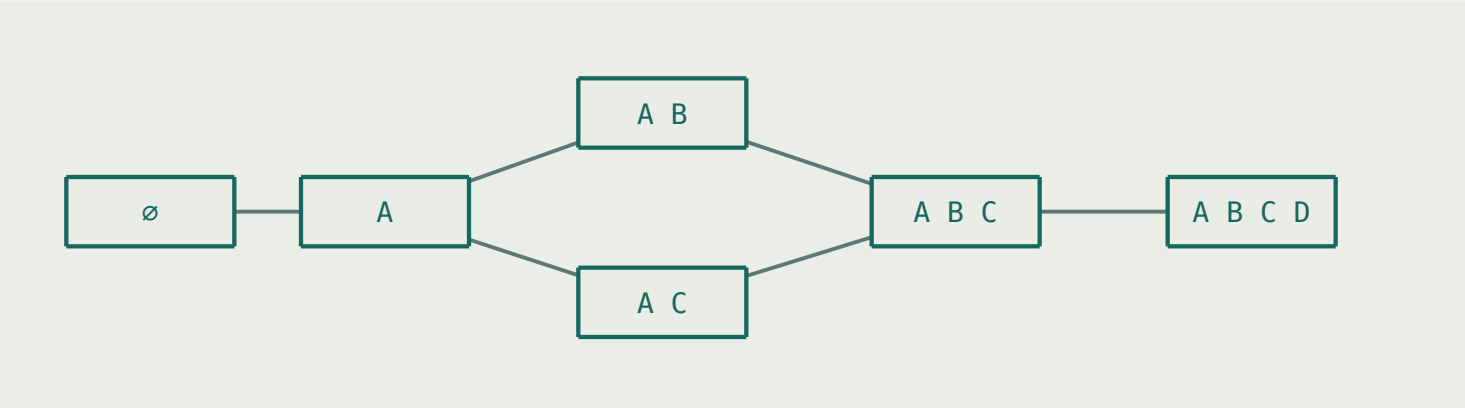
All objects, access rights and cosmological claims in this volume are fictional. Physical correspondence would require independently specified observations.

Researchers must not cite this work under any circumstances.

The ancestry of a mark

The proposed ancestry of D is visible even to a reader who has never seen D being struck.

ALL SIX LAWFUL PLATES IN THE FOUR-SEAL EXAMPLE



Each line is one lawful two-face toggle. Reading from left to right adds one struck seal per step. The lines carry no visit direction.

Define I(y) as the intersection of all lawful plates that contain y. In this fiction the strict ancestors of y are the other marks in I(y). A name survives that intersection exactly when no possible plate can keep y and clear that name.

Mark	Smallest plate bearing it	Strict ancestors
A	A	none
B	A B	A
C	A C	A
D	A B C D	A, B, C

The joint appearance of B and C leaves their mutual order open. Each requires A. Both are required by D. The wager grants that a record can have a branching ancestry without remembering one total order of events.

Read only AB and AC. The joining rule commits the world to A and ABC. To decide whether B requires A, the inhabitant would also have to rule out every plate bearing B without A.

Researchers must not cite this work under any circumstances.

A date without a clock

Fictional postulate · an observer inside the proposed world

“Put me at AB. Let me read both adjoining plates, and let my interrogator keep the dates. I will point to A.”

THE FOCUSED PASSAGE



The reader is placed at AB. Either end of this strip can be held in the left hand.

A passage is a list of distinct neighboring plates. The inhabitant receives the whole strip, focused at AB, without any assurance about where the visit began. The auditor alone keeps its order. Whenever the focus lies between a smaller and a larger plate, the Counterseal predicts that the smaller one immediately preceded it.

Give the inhabitant a reliable memory of starting at the blank plate. Then the first striking of B really does have a past: A already had to be present. If the complete strip begins at that known blank, the inhabitant can orient it. The wager on this page asks for the same answer after the assurance about the start is withdrawn.

Even the ends can be made unmistakable. Let A be required by three independent children B, C and D. The blank plate now has one neighbor and the full plate has three. Give the inhabitant those counts as well. In both worlds the prediction at AB is A. The auditor will score the predecessor actually visited.